| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| **4.** | (a) | (i) | | 80ºC (1)<br>because {the cells are ruptured/ membrane protein denatures} and (all) the pigment has leaked out of the cells/ (1) | | 2 | | 2 | | 2 |
| | | (ii) | | All plots correct = 2 marks<br>1 plot incorrect = 1 mark<br>>1 plot incorrect = 0 marks<br>½ small square tolerance<br>All points plotted correctly with a smooth curve drawn or a dot to dot plot (1)<br>No extrapolation with plot to plot /small amount allowed with line of best fit | | 3 | | 3 | 3 | |
| | | (iii) | | (The absorbance is not directly proportional to the temperature) {as it is not a straight line/ graph is a curve/ a description of changes in gradient} (1) | | | 1 | 1 | | 1 |
| | | (iv) | | **Any two (x1) from:**<br>(Up to 40ºC) As temperature rises the {membrane becomes more fluid/phospholipids gain more kinetic energy/ more gaps form} (1)<br><br>(Above 40ºC) the proteins in the membrane start to denature (1)<br><br>(Above 70ºC) membrane fully destroyed (so all pigment released) (1)<br>reject reference to membrane/ phospholipids denaturing. | | 2 | | 2 | | 2 |

| Question | | | | Marking details | Marks available | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | AO1 | AO2 | AO3 | Total | Maths | Prac |
| | (b) | (i) | | Pigment {leaks/diffuses} {out of the cells / through cell membrane} (1)<br>Acid {denatures the proteins /changes shape of proteins} (1)<br>Ethanol dissolves the {(phospho)lipids/ fatty acids} (1) | | 2 | 1 | 3 | | 3 |
| | | (ii) | | Water had entered the cells / moved in across cell membrane (1)<br>by osmosis (1)<br>as the beetroot cells had a lower water potential than the water (1) | 2 | 1 | | 3 | | 3 |
| | | | | **Question 4 total** | 2 | 10 | 2 | 14 | 3 | 11 |